Chapter 10

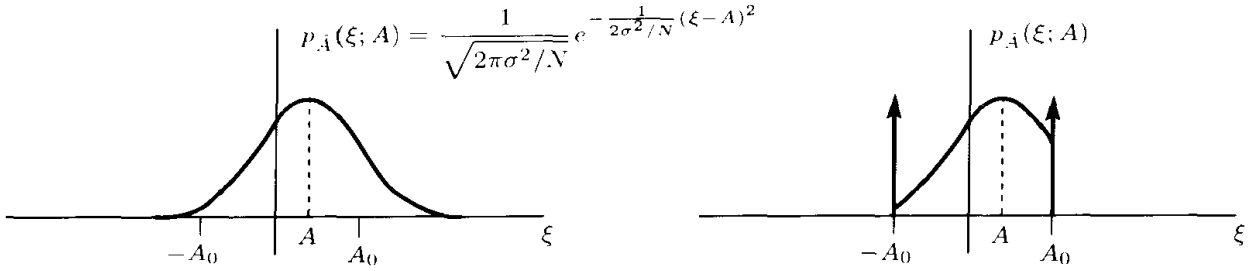
The Bayesian Philosophy

10.1 Introduction

We now depart from the *classical approach* to statistical estimation in which the parameter θ of interest is assumed to be a *deterministic but unknown* constant. Instead, we assume that θ is a *random variable* whose particular *realization* we must estimate. This is the *Bayesian approach*, so named because its implementation is based directly on Bayes' theorem. The motivation for doing so is twofold. First, if we have available some prior knowledge about θ, we can incorporate it into our estimator. The mechanism for doing this requires us to assume that θ is a random variable with a given *prior* PDF. Classical estimation, on the other hand, finds it difficult to make use of any prior knowledge. The Bayesian approach, when applicable, can therefore improve the estimation accuracy. Second, Bayesian estimation is useful in situations where an MVU estimator cannot be found, as for example, when the variance of an unbiased estimator may not be uniformly less than that of all other unbiased estimators. In this instance, it may be true that for *most* values of the parameter an estimator can be found whose mean square error may be less than that of all other estimators. By assigning a PDF to θ we can devise strategies to find that estimator. The resultant estimator can then be said to be optimal "on the average," or with respect to the assumed prior PDF of θ. In this chapter we attempt to motivate the Bayesian approach and to discuss some of the issues surrounding its use. The reader should be aware that this approach to estimation has had a long and controversial history. For a more definitive account [Box and Tiao 1973] is recommended.

10.2 Summary

The Bayesian MSE is defined in (10.2) and is minimized by the estimator of (10.5), which is the mean of the posterior PDF. The example of a DC level in WGN with a Gaussian prior PDF is described in Section 10.4. The minimum MSE estimator for this example is given by (10.11) and represents a weighting between the data knowledge and prior knowledge. The corresponding minimum MSE is given by (10.14). The ability to



(a) PDF of sample mean **(b)** PDF of truncated sample mean

Figure 10.1 Improvement of estimator via prior knowledge

estimate the realization of a random variable based on data is described in Section 10.5 as being due to the correlation between the random variables. Theorem 10.2 summarizes the result that a jointly Gaussian PDF yields a conditional PDF that is also Gaussian, having a mean (10.24) and a covariance (10.25). This is then applied to the Bayesian linear model of (10.26) to yield the posterior PDF as summarized in Theorem 10.3. As will be shown in Chapter 11, the mean of the posterior PDF (10.28) is the minimum MSE estimator for a vector parameter. Section 10.7 discusses nuisance parameters from a Bayesian viewpoint, while Section 10.8 describes the potential difficulties of using a Bayesian estimator in a classical estimation problem.

10.3 Prior Knowledge and Estimation

It is a fundamental rule of estimation theory that the use of prior knowledge will lead to a more accurate estimator. For example, if a parameter is constrained to lie in a known interval, then any good estimator should produce only estimates within that interval. In Example 3.1 it was shown that the MVU estimator of A is the sample mean $\bar{x}$. However, this assumed that A could take on any value in the interval $-\infty < A < \infty$. Due to physical constraints it may be more reasonable to assume that A can take on only values in the finite interval $-A_0 \leq A \leq A_0$. To retain $\hat{A} = \bar{x}$ as the best estimator would be undesirable since $\hat{A}$ may yield values outside the known interval. As shown in Figure 10.1a, this is due to noise effects. Certainly, we would expect to improve our estimation if we used the *truncated* sample mean estimator

$$\check{A} = \begin{cases} -A_0 & \bar{x} < -A_0 \\ \bar{x} & -A_0 \leq \bar{x} \leq A_0 \\ A_0 & \bar{x} > A_0 \end{cases}$$

which would be consistent with the known constraints. Such an estimator would have the PDF

$$\begin{aligned} p_{\check{A}}(\xi; A) &= \Pr\{\bar{x} \leq -A_0\}\delta(\xi + A_0) \\ &\quad + p_{\hat{A}}(\xi; A)[u(\xi + A_0) - u(\xi - A_0)] \\ &\quad + \Pr\{\bar{x} \geq A_0\}\delta(\xi - A_0) \end{aligned} \tag{10.1}$$

where $u(x)$ is the unit step function. This is shown in Figure 10.1b. It is seen that $\check{A}$ is a biased estimator. However, if we compare the MSE of the two estimators, we note that for any A in the interval $-A_0 \leq A \leq A_0$

$$\begin{aligned}
\text{mse}(\hat{A}) &= \int_{-\infty}^{\infty} (\xi - A)^2 p_{\hat{A}}(\xi; A)\, d\xi \\
&= \int_{-\infty}^{-A_0} (\xi - A)^2 p_{\hat{A}}(\xi; A)\, d\xi + \int_{-A_0}^{A_0} (\xi - A)^2 p_{\hat{A}}(\xi; A)\, d\xi \\
&\quad + \int_{A_0}^{\infty} (\xi - A)^2 p_{\hat{A}}(\xi; A)\, d\xi \\
&> \int_{-\infty}^{-A_0} (-A_0 - A)^2 p_{\hat{A}}(\xi; A)\, d\xi + \int_{-A_0}^{A_0} (\xi - A)^2 p_{\hat{A}}(\xi; A)\, d\xi \\
&\quad + \int_{A_0}^{\infty} (A_0 - A)^2 p_{\hat{A}}(\xi; A)\, d\xi \\
&= \text{mse}(\check{A}).
\end{aligned}$$

Hence, $\check{A}$, the truncated sample mean estimator, is better than the sample mean estimator in terms of MSE. Although $\hat{A}$ is still the MVU estimator, we have been able to reduce the *mean square error* by allowing the estimator to be biased. In as much as we have been able to produce a better estimator, the question arises as to whether an *optimal* estimator exists for this problem. (The reader may recall that in the classical case the MSE criterion of optimality usually led to unrealizable estimators. We shall see that this is not a problem in the Bayesian approach.) We can answer affirmatively but only after reformulating the data model. Knowing that A must lie in a known interval, we suppose that the true value of A has been chosen from that interval. We then model the process of choosing a value as a random event to which a PDF can be assigned. With knowledge only of the interval and no inclination as to whether A should be nearer any particular value, it makes sense to assign a $\mathcal{U}[-A_0, A_0]$ PDF to the *random variable* A. The overall data model then appears as in Figure 10.2. As shown there, the act of choosing A according to the given PDF represents the departure of the Bayesian approach from the classical approach. The problem, as always, is to estimate the *value* of A or the *realization* of the random variable. However, now we can incorporate our knowledge of *how* A was chosen. For example, we might attempt to find an estimator $\hat{A}$ that would minimize the *Bayesian* MSE defined as

$$\text{Bmse}(\hat{A}) = E[(A - \hat{A})^2]. \tag{10.2}$$

We choose to define the error as $A - \hat{A}$ in contrast to the classical estimation error of $\hat{A} - A$. This definition will be useful later when we discuss a vector space interpretation of the Bayesian estimator. In (10.2) we emphasize that since A is a random variable, the expectation operator is with respect to the *joint PDF* $p(\mathbf{x}, A)$. This is a fundamentally different MSE than in the classical case. We distinguish it by using the Bmse notation. To appreciate the difference compare the classical MSE

$$\text{mse}(\hat{A}) = \int (\hat{A} - A)^2 p(\mathbf{x}; A)\, d\mathbf{x} \tag{10.3}$$

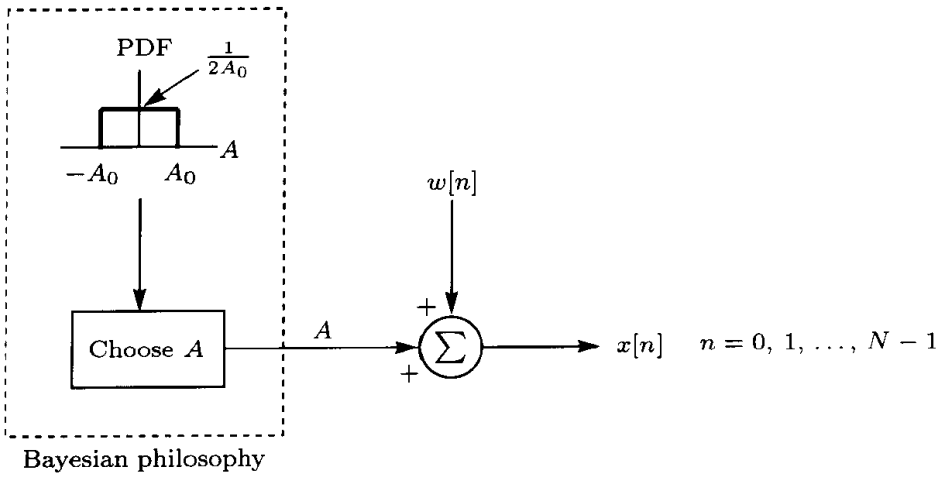


Figure 10.2 Bayesian approach to data modeling

to the Bayesian MSE

$$\text{Bmse}(\hat{A}) = \int\int (A - \hat{A})^2 p(\mathbf{x}, A)\, d\mathbf{x}\, dA. \tag{10.4}$$

Even the underlying experiments are different, as is reflected in the averaging PDFs. If we were to assess the MSE performance using a Monte Carlo computer simulation, then for the classical approach we would choose a realization of $w[n]$ and add it to a *given* A. This procedure would be repeated M times. Each time we would add a new realization of $w[n]$ to the *same* A. In the Bayesian approach, for each realization we would choose A according to its PDF $\mathcal{U}[-A_0, A_0]$ and then generate $w[n]$ (assuming that $w[n]$ is independent of A). We would then repeat this procedure M times. In the classical case we would obtain a MSE for each assumed value of A, while in the Bayesian case the single MSE figure obtained would be an average over the PDF of A. Note that whereas the classical MSE will depend on A, and hence estimators that attempt to minimize the MSE will usually depend on A (see Section 2.4), the Bayesian MSE will not. In effect, we have integrated the parameter dependence away! It should be clear that comparing classical and Bayesian estimators is like comparing "apples and oranges." The reader who is tempted to do so may become thoroughly confused. Nonetheless, at times, the *forms* of the estimators will be identical (see Problem 10.1).

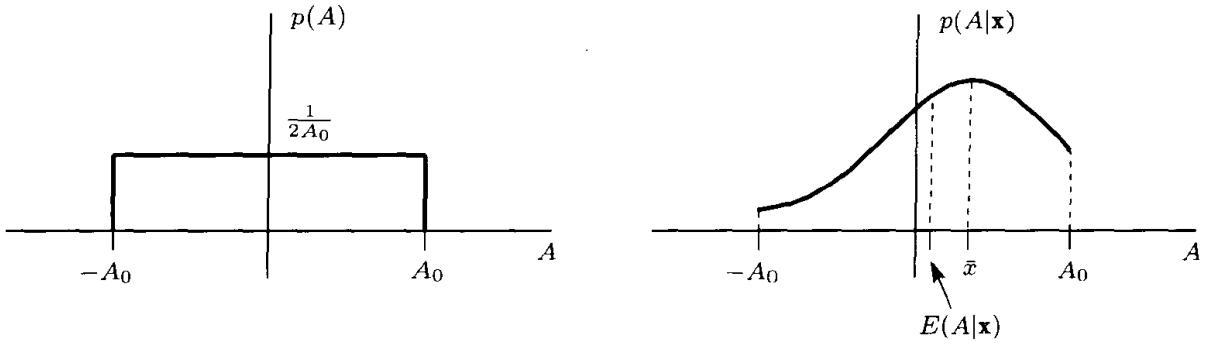
To complete our example we now derive the estimator that minimizes the Bayesian MSE. First, we use Bayes' theorem to write

$$p(\mathbf{x}, A) = p(A|\mathbf{x})p(\mathbf{x})$$

so that

$$\text{Bmse}(\hat{A}) = \int \left[\int (A - \hat{A})^2 p(A|\mathbf{x})\, dA \right] p(\mathbf{x})\, d\mathbf{x}.$$

Now since $p(\mathbf{x}) \geq 0$ for all $\mathbf{x}$, if the integral in brackets can be minimized for each $\mathbf{x}$, then the Bayesian MSE will be minimized. Hence, fixing $\mathbf{x}$ so that $\hat{A}$ is a scalar variable



(a) Prior PDF (b) Posterior PDF

Figure 10.3 Comparison of prior and posterior PDFs

(as opposed to a general function of $\mathbf{x}$), we have

$$\begin{aligned}\frac{\partial}{\partial \hat{A}} \int (A-\hat{A})^2 p(A|\mathbf{x})\, dA &= \int \frac{\partial}{\partial \hat{A}} (A-\hat{A})^2 p(A|\mathbf{x})\, dA \\ &= \int -2(A-\hat{A}) p(A|\mathbf{x})\, dA \\ &= -2\int A p(A|\mathbf{x})\, dA + 2\hat{A}\int p(A|\mathbf{x})\, dA\end{aligned}$$

which when set equal to zero results in

$$\hat{A} = \int A p(A|\mathbf{x})\, dA$$

or finally

$$\hat{A} = E(A|\mathbf{x}) \tag{10.5}$$

since the conditional PDF must integrate to 1. It is seen that the optimal estimator in terms of minimizing the Bayesian MSE is the *mean* of the *posterior* PDF $p(A|\mathbf{x})$ (see also Problem 10.5 for an alternative derivation). The posterior PDF refers to the PDF of A *after* the data have been observed. In contrast, $p(A)$ or

$$p(A) = \int p(\mathbf{x}, A)\, d\mathbf{x}$$

may be thought of as the prior PDF of A, indicating the PDF *before* the data are observed. We will henceforth term the estimator that minimizes the Bayesian MSE the minimum mean square error (MMSE) estimator. Intuitively, the effect of observing data will be to concentrate the PDF of A as shown in Figure 10.3 (see also Problem 10.15). This is because knowledge of the data should reduce our uncertainty about A. We will return to this idea later.

In determining the MMSE estimator we first require the posterior PDF. We can use Bayes' rule to determine it as

$$\begin{aligned} p(A|\mathbf{x}) &= \frac{p(\mathbf{x}|A)p(A)}{p(\mathbf{x})} \\ &= \frac{p(\mathbf{x}|A)p(A)}{\int p(\mathbf{x}|A)p(A)\,dA}. \end{aligned} \tag{10.6}$$

Note that the denominator is just a normalizing factor, independent of A, needed to ensure that $p(A|\mathbf{x})$ integrates to 1. If we continue our example, we recall that the prior PDF $p(A)$ is $\mathcal{U}[-A_0, A_0]$. To specify the conditional PDF $p(\mathbf{x}|A)$ we need to further assume that the choice of A via $p(A)$ does not affect the PDF of the noise samples or that $w[n]$ is independent of A. Then, for $n = 0, 1, \ldots, N-1$

$$\begin{aligned} p_x(x[n]|A) &= p_w(x[n] - A|A) \\ &= p_w(x[n] - A) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2}(x[n] - A)^2\right] \end{aligned}$$

and therefore

$$p(\mathbf{x}|A) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - A)^2\right]. \tag{10.7}$$

It is apparent that the PDF is identical *in form* to the usual classical PDF $p(\mathbf{x}; A)$. In the Bayesian case, however, the PDF is a *conditional* PDF, hence the "|" separator, while in the classical case, it represents an unconditional PDF, albeit parameterized by A, hence the separator ";" (see also Problem 10.6). Using (10.6) and (10.7), the posterior PDF becomes

$$p(A|\mathbf{x}) = \begin{cases} \dfrac{\dfrac{1}{2A_0(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\dfrac{1}{2\sigma^2}\displaystyle\sum_{n=0}^{N-1}(x[n] - A)^2\right]}{\displaystyle\int_{-A_0}^{A_0} \frac{1}{2A_0(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - A)^2\right] dA} & |A| \leq A_0 \\ 0 & |A| > A_0. \end{cases}$$

But

$$\begin{aligned} \sum_{n=0}^{N-1}(x[n] - A)^2 &= \sum_{n=0}^{N-1} x^2[n] - 2NA\bar{x} + NA^2 \\ &= N(A - \bar{x})^2 + \sum_{n=0}^{N-1} x^2[n] - N\bar{x}^2 \end{aligned}$$

so that we have

$$p(A|\mathbf{x}) = \begin{cases} \frac{1}{c\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left[-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right] & |A| \leq A_0 \\ 0 & |A| > A_0 \end{cases} \tag{10.8}$$

The factor c is determined by the requirement that $p(A|\mathbf{x})$ integrate to 1, resulting in

$$c = \int_{-A_0}^{A_0} \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left[-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right] dA.$$

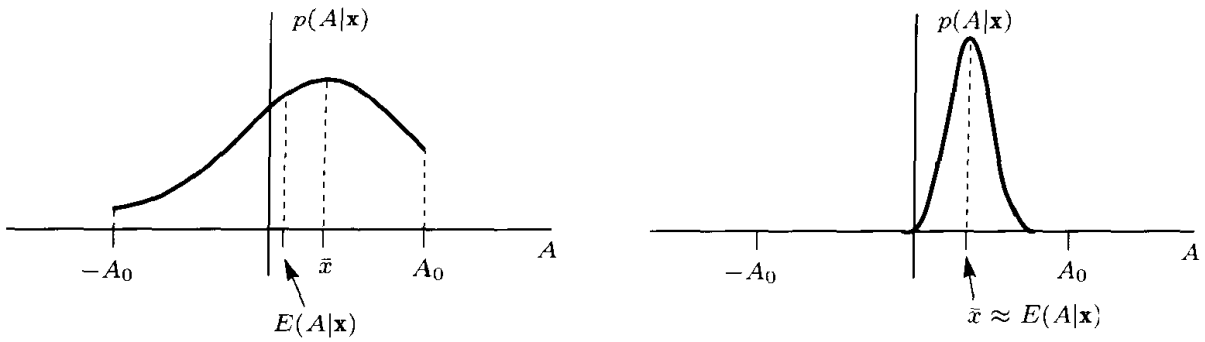
The PDF is seen to be a truncated Gaussian, as shown in Figure 10.3b. The MMSE estimator, which is the mean of $p(A|\mathbf{x})$, is

$$\begin{aligned} \hat{A} &= E(A|\mathbf{x}) \\ &= \int_{-\infty}^{\infty} Ap(A|\mathbf{x})\, dA \\ &= \frac{\int_{-A_0}^{A_0} A\frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left[-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right] dA}{\int_{-A_0}^{A_0} \frac{1}{\sqrt{2\pi\frac{\sigma^2}{N}}} \exp\left[-\frac{1}{2\frac{\sigma^2}{N}}(A-\bar{x})^2\right] dA}. \end{aligned} \tag{10.9}$$

Although this cannot be evaluated in closed form, we note that $\hat{A}$ will be a function of $\bar{x}$ as well as of A_0 and σ^2 (see Problem 10.7). The MMSE estimator will not be $\bar{x}$ due to the truncation shown in Figure 10.3b unless A_0 is so large that there is effectively no truncation. This will occur if $A_0 \gg \sqrt{\sigma^2/N}$. Otherwise, the estimator will be "biased" towards zero as opposed to being equal to $\bar{x}$. This is because the prior knowledge embodied in $p(A)$ would in the absence of the data $\mathbf{x}$ produce the MMSE estimator (see Problem 10.8)

$$\hat{A} = E(A) = 0.$$

The effect of the data is to position the posterior mean between $A = 0$ and $A = \bar{x}$ in a compromise between the prior knowledge and that contributed by the data. To further appreciate this weighting consider what happens as N becomes large so that the data knowledge becomes more important. As shown in Figure 10.4, as N increases, we have from (10.8) that the posterior PDF becomes more concentrated about $\bar{x}$ (since σ^2/N decreases). Hence, it becomes nearly Gaussian, and its mean becomes just $\bar{x}$. The MMSE estimator relies less and less on the prior knowledge and more on the data. It is said that the data "swamps out" the prior knowledge.



(a) Short data record **(b)** Large data record

Figure 10.4 Effect of increasing data record on posterior PDF

The results of this example are true in general and are now summarized. The Bayesian approach to parameter estimation assumes that the parameter to be estimated is a realization of the random variable θ. As such, we assign a prior PDF $p(\theta)$ to it. After the data are observed, our state of knowledge about the parameter is summarized by the posterior PDF $p(\theta|\mathbf{x})$. An optimal estimator is defined to be the one that minimizes the MSE when averaged over all realizations of θ and $\mathbf{x}$, the so-called Bayesian MSE. This estimator is the mean of the posterior PDF or $\hat{\theta} = E(\theta|\mathbf{x})$. The estimator is determined explicitly as

$$\hat{\theta} = E(\theta|\mathbf{x}) = \int \theta p(\theta|\mathbf{x})\, d\theta. \tag{10.10}$$

The MMSE estimator will in general depend on the prior knowledge as well as the data. If the prior knowledge is weak relative to that of the data, then the estimator will ignore the prior knowledge. Otherwise, the estimator will be "biased" towards the prior mean. As expected, the use of prior information always improves the estimation accuracy (see Example 10.1).

The choice of a prior PDF is critical in Bayesian estimation. The wrong choice will result in a poor estimator, similar to the problems of a classical estimator designed with an incorrect data model. Much of the controversy surrounding the use of Bayesian estimators stems from the inability in practice to be able to justify the prior PDF. Suffice it to say that unless the prior PDF can be based on the physical constraints of the problem, then classical estimation is more appropriate.

10.4 Choosing a Prior PDF

As shown in the previous section, once a prior PDF has been chosen, the MMSE estimator follows directly from (10.10). There is no question of existence as there is with the MVU estimator in the classical approach. The only practical stumbling block that remains, however, is whether or not $E(\theta|\mathbf{x})$ can be determined in closed form. In

the introductory example the posterior PDF $p(A|\mathbf{x})$ as given by (10.8) could not be found explicitly due to the need to normalize $p(\mathbf{x}|A)p(A)$ so that it integrates to 1. Additionally, the posterior mean could not be found, as evidenced by (10.9). We would have to resort to numerical integration to actually implement the MMSE estimator. This problem is compounded considerably in the vector parameter case. There the posterior PDF becomes

$$p(\boldsymbol{\theta}|\mathbf{x}) = \frac{p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})\,d\boldsymbol{\theta}}$$

which requires a p-dimensional integration over $\boldsymbol{\theta}$. Additionally, the mean needs to be evaluated (see Chapter 11), requiring further integration. For practical MMSE estimators we need to be able to express them in closed form. The next example illustrates an important case where this is possible.

Example 10.1 - DC Level in WGN - Gaussian Prior PDF

We now modify our prior knowledge for the introductory example. Instead of assuming the uniform prior PDF

$$p(A) = \begin{cases} \frac{1}{2A_0} & |A| \leq A_0 \\ 0 & |A| > A_0 \end{cases}$$

which led to an intractable integration, consider the Gaussian prior PDF

$$p(A) = \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp\left[-\frac{1}{2\sigma_A^2}(A-\mu_A)^2\right].$$

The two prior PDFs clearly express different prior knowledge about A, although with $\mu_A = 0$ and $3\sigma_A = A_0$ the Gaussian prior PDF could be thought of as incorporating the knowledge that $|A| \leq A_0$. Of course, values of A near zero are thought to be more probable with the Gaussian prior PDF. Now if

$$\begin{aligned} p(\mathbf{x}|A) &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right] \\ &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]\right] \exp\left[-\frac{1}{2\sigma^2}(NA^2 - 2NA\bar{x})\right] \end{aligned}$$

we have

$$
\begin{aligned}
p(A|\mathbf{x}) &= \frac{p(\mathbf{x}|A)p(A)}{\int p(\mathbf{x}|A)p(A)\,dA} \\
&= \frac{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}\sqrt{2\pi\sigma_A^2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1} x^2[n]\right] \exp\left[-\frac{1}{2\sigma^2}(NA^2 - 2NA\bar{x})\right]}{\int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}\sqrt{2\pi\sigma_A^2}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1} x^2[n]\right] \exp\left[-\frac{1}{2\sigma^2}(NA^2 - 2NA\bar{x})\right]} \\
&\qquad \cdot \frac{\exp\left[-\frac{1}{2\sigma_A^2}(A-\mu_A)^2\right]}{\exp\left[-\frac{1}{2\sigma_A^2}(A-\mu_A)^2\right] dA} \\
&= \frac{\exp\left[-\frac{1}{2}\left(\frac{1}{\sigma^2}(NA^2 - 2NA\bar{x}) + \frac{1}{\sigma_A^2}(A-\mu_A)^2\right)\right]}{\int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{1}{\sigma^2}(NA^2 - 2NA\bar{x}) + \frac{1}{\sigma_A^2}(A-\mu_A)^2\right)\right] dA} \\
&= \frac{\exp\left[-\frac{1}{2}Q(A)\right]}{\int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}Q(A)\right] dA}.
\end{aligned}
$$

Note, however, that the denominator does not depend on A, being a normalizing factor, and the argument of the exponential is quadratic in A. Hence, $p(A|\mathbf{x})$ must be a Gaussian PDF whose mean and variance depend on $\mathbf{x}$. Continuing, we have for $Q(A)$

$$
\begin{aligned}
Q(A) &= \frac{N}{\sigma^2}A^2 - \frac{2NA\bar{x}}{\sigma^2} + \frac{A^2}{\sigma_A^2} - \frac{2\mu_A A}{\sigma_A^2} + \frac{\mu_A^2}{\sigma_A^2} \\
&= \left(\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}\right)A^2 - 2\left(\frac{N}{\sigma^2}\bar{x} + \frac{\mu_A}{\sigma_A^2}\right)A + \frac{\mu_A^2}{\sigma_A^2}.
\end{aligned}
$$

Let

$$
\begin{aligned}
\sigma_{A|x}^2 &= \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}} \\
\mu_{A|x} &= \left(\frac{N}{\sigma^2}\bar{x} + \frac{\mu_A}{\sigma_A^2}\right)\sigma_{A|x}^2.
\end{aligned}
$$

Then, by completing the square we have

$$
Q(A) = \frac{1}{\sigma_{A|x}^2}\left(A^2 - 2\mu_{A|x}A + \mu_{A|x}^2\right) - \frac{\mu_{A|x}^2}{\sigma_{A|x}^2} + \frac{\mu_A^2}{\sigma_A^2}
$$

$$= \frac{1}{\sigma_{A|x}^2}(A-\mu_{A|x})^2 - \frac{\mu_{A|x}^2}{\sigma_{A|x}^2} + \frac{\mu_A^2}{\sigma_A^2}$$

so that

$$p(A|\mathbf{x}) = \frac{\exp\left[-\frac{1}{2\sigma_{A|x}^2}(A-\mu_{A|x})^2\right]\exp\left[-\frac{1}{2}\left(\frac{\mu_A^2}{\sigma_A^2}-\frac{\mu_{A|x}^2}{\sigma_{A|x}^2}\right)\right]}{\int_{-\infty}^{\infty}\exp\left[-\frac{1}{2\sigma_{A|x}^2}(A-\mu_{A|x})^2\right]\exp\left[-\frac{1}{2}\left(\frac{\mu_A^2}{\sigma_A^2}-\frac{\mu_{A|x}^2}{\sigma_{A|x}^2}\right)\right]dA}$$

$$= \frac{1}{\sqrt{2\pi\sigma_{A|x}^2}}\exp\left[-\frac{1}{2\sigma_{A|x}^2}(A-\mu_{A|x})^2\right]$$

where the last step follows from the requirement that $p(A|\mathbf{x})$ integrate to 1. The posterior PDF is also Gaussian, as claimed. (This result could also have been obtained by using Theorem 10.2 since $A, \mathbf{x}$ are jointly Gaussian.) In this form the MMSE estimator is readily found as

$$\begin{aligned}\hat{A} &= E(A|\mathbf{x}) \\ &= \mu_{A|x} \\ &= \frac{\frac{N}{\sigma^2}\bar{x} + \frac{\mu_A}{\sigma_A^2}}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}}\end{aligned}$$

or finally, the MMSE estimator is

$$\begin{aligned}\hat{A} &= \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}\bar{x} + \frac{\frac{\sigma^2}{N}}{\sigma_A^2 + \frac{\sigma^2}{N}}\mu_A \\ &= \alpha\bar{x} + (1-\alpha)\mu_A \end{aligned} \tag{10.11}$$

where

$$\alpha = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}.$$

Note that α is a weighting factor since $0 < \alpha < 1$. Using a Gaussian prior PDF, we are able to determine the MMSE estimator explicitly. It is interesting to examine the interplay between the prior knowledge and the data. When there is little data so that $\sigma_A^2 \ll \sigma^2/N$, α is small and $\hat{A} \approx \mu_A$, but as more data are observed so that $\sigma_A^2 \gg \sigma^2/N$, $\alpha \approx 1$ and $\hat{A} \approx \bar{x}$. The weighting factor α depends directly on our confidence in the prior knowledge or σ_A^2 and the data knowledge or σ^2/N. (The quantity σ^2/N is interpreted as the *conditional* variance or $E[(\bar{x}-A)^2|A]$). Alternatively, we may view this process by examining the posterior PDF as N increases. Referring to Figure 10.5, as the data record length N increases, the posterior PDF becomes narrower. This is because the

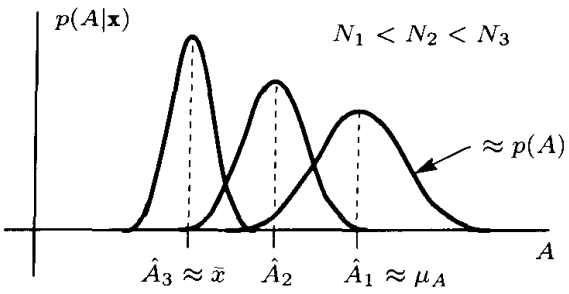


Figure 10.5 Effect of increasing data record length on posterior PDF

posterior variance

$$\text{var}(A|\mathbf{x}) = \sigma^2_{A|x} = \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}} \tag{10.12}$$

will decrease. Also, the posterior mean (10.11) or $\hat{A}$ will also change with increasing N. For small N it will be approximately μ_A but will approach $\bar{x}$ for increasing N. In fact, as $N \to \infty$, we will have $\hat{A} \to \bar{x}$, which in turn approaches the true value of A chosen. Observe that if there is no prior knowledge, which can be modeled by letting $\sigma_A^2 \to \infty$, then $\hat{A} \to \bar{x}$ for any data record length. The "classical" estimator is obtained. Finally, it was originally claimed that by using prior knowledge we could improve the estimation accuracy. To see why this is so recall that

$$\text{Bmse}(\hat{A}) = E[(A - \hat{A})^2]$$

where we evaluate the expectation with respect to $p(\mathbf{x}, A)$. But

$$\begin{aligned} \text{Bmse}(\hat{A}) &= \int\int (A - \hat{A})^2 p(\mathbf{x}, A)\, d\mathbf{x}\, dA \\ &= \int\int (A - \hat{A})^2 p(A|\mathbf{x})\, dA p(\mathbf{x})\, d\mathbf{x}. \end{aligned}$$

Since $\hat{A} = E(A|\mathbf{x})$, we have

$$\begin{aligned} \text{Bmse}(\hat{A}) &= \int\int [A - E(A|\mathbf{x})]^2 p(A|\mathbf{x})\, dA p(\mathbf{x})\, d\mathbf{x} \\ &= \int \text{var}(A|\mathbf{x}) p(\mathbf{x})\, d\mathbf{x}. \end{aligned} \tag{10.13}$$

We see that *the Bayesian MSE is just the variance of the posterior PDF when averaged over the PDF of* $\mathbf{x}$. As such, we have

$$\begin{aligned} \text{Bmse}(\hat{A}) &= \int \sigma^2_{A|x} p(\mathbf{x})\, d\mathbf{x} \\ &= \frac{1}{\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2}} \end{aligned}$$

since $\sigma^2_{A|x}$ does not depend on $\mathbf{x}$. This can be rewritten as

$$\text{Bmse}(\hat{A}) = \frac{\sigma^2}{N}\left(\frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}\right) \tag{10.14}$$

so that finally we see that

$$\text{Bmse}(\hat{A}) < \frac{\sigma^2}{N}$$

where σ^2/N is the minimum MSE obtained when no prior knowledge is available (let $\sigma_A^2 \to \infty$). Clearly, any prior knowledge *when modeled in the Bayesian sense* will improve our *Bayesian estimator.* ◇

Gaussian prior PDFs are quite useful in practice due to their mathematical tractability, as illustrated by the previous example. The basic property that makes this so is the *reproducing property.* If $p(\mathbf{x}, A)$ is Gaussian, then $p(A)$, being the marginal PDF, is Gaussian, as is the posterior PDF $p(A|\mathbf{x})$. Hence, the form of the PDF remains the same, as it is conditioned on $\mathbf{x}$. Only the mean and variance change. Another example of a PDF sharing this property is given in Problem 10.10. Furthermore, Gaussian prior PDFs occur naturally in many practical problems. For the previous example we can envision the problem of measuring the DC voltage of a power source by means of a DC voltmeter. If we set the power source to 10 volts, for example, we would probably be willing to assume that the true voltage is close to 10 volts. Our prior knowledge then might be modeled as $A \sim \mathcal{N}(10, \sigma_A^2)$, where σ_A^2 would be small for a precision power source and large for a less reliable one. Next, we could take N measurements of the voltage. Our model for the measurements could be $x[n] = A + w[n]$, where the voltmeter error $w[n]$ is modeled by WGN with variance σ^2. The value of σ^2 would reflect our confidence in the quality of the voltmeter. The MMSE estimator of the true voltage would be given by (10.11). If we repeated the procedure for an ensemble of power sources and voltmeters with the same error characteristics, then our estimator would minimize the Bayesian MSE.

10.5 Properties of the Gaussian PDF

We now generalize the results of the previous section by examining the properties of the Gaussian PDF. The results of this section will be needed for the derivations of Bayesian estimators in the next chapter. The bivariate Gaussian PDF is first investigated to illustrate the important properties. Then, the corresponding results for the general multivariate Gaussian PDF are described. The remarkable property that we shall exploit is that the posterior PDF is also Gaussian, although with a different mean and variance. Some physical interpretations are stressed.

Consider a jointly Gaussian random vector $[x\, y]^T$ whose PDF is

$$p(x,y) = \frac{1}{2\pi \det^{\frac{1}{2}}(\mathbf{C})} \exp\left[-\frac{1}{2}\begin{bmatrix} x - E(x) \\ y - E(y) \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} x - E(x) \\ y - E(y) \end{bmatrix}\right]. \tag{10.15}$$

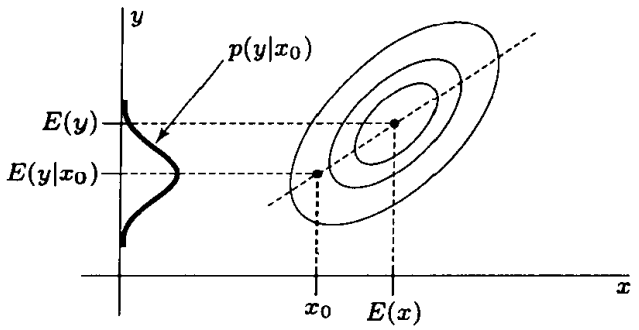


Figure 10.6 Contours of constant density for bivariate Gaussian PDF

This is also termed the *bivariate* Gaussian PDF. The mean vector and covariance matrix are

$$E\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} E(x) \\ E(y) \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} \text{var}(x) & \text{cov}(x,y) \\ \text{cov}(y,x) & \text{var}(y) \end{bmatrix}.$$

Note that the marginal PDFs $p(x)$ and $p(y)$ are also Gaussian, as can be verified by the integrations

$$p(x) = \int_{-\infty}^{\infty} p(x,y)\,dy = \frac{1}{\sqrt{2\pi \text{var}(x)}} \exp\left[-\frac{1}{2\text{var}(x)}(x - E(x))^2\right]$$

$$p(y) = \int_{-\infty}^{\infty} p(x,y)\,dx = \frac{1}{\sqrt{2\pi \text{var}(y)}} \exp\left[-\frac{1}{2\text{var}(y)}(y - E(y))^2\right].$$

The contours along which the PDF $p(x,y)$ is constant are those values of x and y for which

$$\begin{bmatrix} x - E(x) \\ y - E(y) \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} x - E(x) \\ y - E(y) \end{bmatrix}$$

is a constant. They are shown in Figure 10.6 as elliptical contours. Once x, say x_0, is observed, the conditional PDF of y becomes

$$p(y|x_0) = \frac{p(x_0,y)}{p(x_0)} = \frac{p(x_0,y)}{\int_{-\infty}^{\infty} p(x_0,y)\,dy}$$

so that the conditional PDF of y is that of the cross section shown in Figure 10.6 when suitably normalized to integrate to 1. It is readily seen that since $p(x_0,y)$ (where x_0 is a fixed number) has the Gaussian form in y (from (10.15) the exponential argument is quadratic in y), the conditional PDF must also be Gaussian. Since $p(y)$ is also Gaussian, we may view this property as saying that if x and y are jointly Gaussian, the prior PDF $p(y)$ and posterior PDF $p(y|x)$ are both Gaussian. In Appendix 10A we derive the exact PDF as summarized in the following theorem.

Theorem 10.1 (Conditional PDF of Bivariate Gaussian) *If x and y are distributed according to a bivariate Gaussian PDF with mean vector $[E(x)\,E(y)]^T$ and covariance matrix*

$$\mathbf{C} = \begin{bmatrix} \text{var}(x) & \text{cov}(x,y) \\ \text{cov}(y,x) & \text{var}(y) \end{bmatrix}$$

so that

$$p(x,y) = \frac{1}{2\pi \det^{\frac{1}{2}}(\mathbf{C})} \exp\left[-\frac{1}{2}\begin{bmatrix} x-E(x) \\ y-E(y) \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} x-E(x) \\ y-E(y) \end{bmatrix}\right],$$

then the conditional PDF $p(y|x)$ is also Gaussian and

$$E(y|x) = E(y) + \frac{\text{cov}(x,y)}{\text{var}(x)}(x - E(x)) \tag{10.16}$$

$$\text{var}(y|x) = \text{var}(y) - \frac{\text{cov}^2(x,y)}{\text{var}(x)}. \tag{10.17}$$

We can view this result in the following way. Before observing x, the random variable y is distributed according to the prior PDF $p(y)$ or $y \sim \mathcal{N}(E(y), \text{var}(y))$. After observing x, the random variable y is distributed according to the posterior PDF $p(y|x)$ given in Theorem 10.1. Only the mean and variance have changed. Assuming that x and y are not independent and hence $\text{cov}(x,y) \neq 0$, the posterior PDF becomes more concentrated since there is less uncertainty about y. To verify this, note from (10.17) that

$$\begin{aligned} \text{var}(y|x) &= \text{var}(y)\left[1 - \frac{\text{cov}^2(x,y)}{\text{var}(x)\text{var}(y)}\right] \\ &= \text{var}(y)(1-\rho^2) \end{aligned} \tag{10.18}$$

where

$$\rho = \frac{\text{cov}(x,y)}{\sqrt{\text{var}(x)\text{var}(y)}} \tag{10.19}$$

is the correlation coefficient satisfying $|\rho| \leq 1$. From our previous discussions we also realize that $E(y|x)$ is the MMSE estimator of y after observing x, so that from (10.16)

$$\hat{y} = E(y) + \frac{\text{cov}(x,y)}{\text{var}(x)}(x - E(x)). \tag{10.20}$$

In normalized form (a random variable with zero mean and unity variance) this becomes

$$\frac{\hat{y} - E(y)}{\sqrt{\text{var}(y)}} = \frac{\text{cov}(x,y)}{\sqrt{\text{var}(x)\text{var}(y)}} \frac{x - E(x)}{\sqrt{\text{var}(x)}}$$

or

$$\hat{y}_n = \rho x_n. \tag{10.21}$$

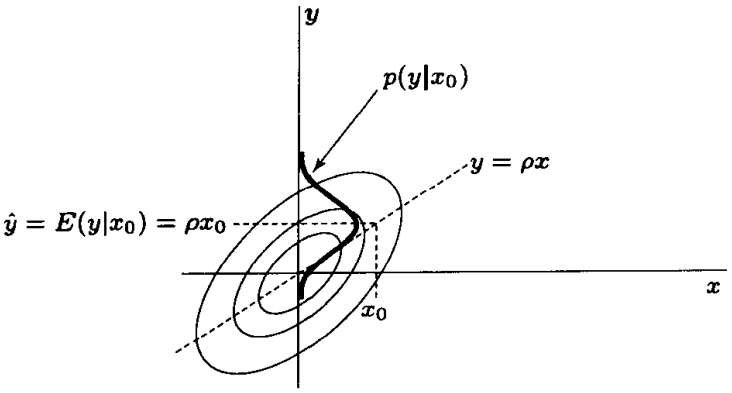


Figure 10.7
Contours of constant density of normalized bivariate PDF

The correlation coefficient then acts to scale the normalized observation x_n to obtain the MMSE estimator of the normalized realization of the random variable y_n. If the random variables are already normalized ($E(x) = E(y) = 0$, $\text{var}(x) = \text{var}(y) = 1$), the constant PDF contours appear as in Figure 10.7. The locations of the peaks of $p(x, y)$, when considered as a function of y for each x, is the dashed line $y = \rho x$, and it is readily shown that $\hat{y} = E(y|x) = \rho x$ (see Problem 10.12). *The MMSE estimator therefore exploits the correlation between the random variables to estimate the realization of one based on the realization of the other.*

The minimum MSE is, from (10.13) and (10.18),

$$\begin{aligned} \text{Bmse}(\hat{y}) &= \int \text{var}(y|x)p(x)\,dx \\ &= \text{var}(y|x) \\ &= \text{var}(y)(1-\rho^2) \end{aligned} \tag{10.22}$$

since the posterior variance does not depend on x ($\text{var}(y)$ and ρ depend on the covariance matrix only). Hence, the quality of our estimator also depends on the correlation coefficient, which is a measure of the statistical dependence between x and y.

To generalize these results consider a jointly Gaussian vector $[\mathbf{x}^T\ \mathbf{y}^T]^T$, where $\mathbf{x}$ is $k \times 1$ and $\mathbf{y}$ is $l \times 1$. In other words, $[\mathbf{x}^T\ \mathbf{y}^T]^T$ is distributed according to a multivariate Gaussian PDF. Then, the conditional PDF of $\mathbf{y}$ for a given $\mathbf{x}$ is also Gaussian, as summarized in the following theorem (see Appendix 10A for proof).

Theorem 10.2 (Conditional PDF of Multivariate Gaussian) *If* $\mathbf{x}$ *and* $\mathbf{y}$ *are jointly Gaussian, where* $\mathbf{x}$ *is* $k \times 1$ *and* $\mathbf{y}$ *is* $l \times 1$*, with mean vector* $[E(\mathbf{x})^T\ E(\mathbf{y})^T]^T$ *and partitioned covariance matrix*

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{xx} & \mathbf{C}_{xy} \\ \mathbf{C}_{yx} & \mathbf{C}_{yy} \end{bmatrix} = \begin{bmatrix} k \times k & k \times l \\ l \times k & l \times l \end{bmatrix} \tag{10.23}$$

so that

$$p(\mathbf{x}, \mathbf{y}) = \frac{1}{(2\pi)^{\frac{k+l}{2}} \det^{\frac{1}{2}}(\mathbf{C})} \exp\left[-\frac{1}{2}\left(\begin{bmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{bmatrix}\right)^T \mathbf{C}^{-1} \left(\begin{bmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{bmatrix}\right)\right],$$

then the conditional PDF $p(\mathbf{y}|\mathbf{x})$ *is also Gaussian and*

$$E(\mathbf{y}|\mathbf{x}) = E(\mathbf{y}) + \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \quad (10.24)$$

$$\mathbf{C}_{y|x} = \mathbf{C}_{yy} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy}. \quad (10.25)$$

Note that the covariance matrix of the conditional PDF does not depend on $\mathbf{x}$, although this property is not generally true. This will be useful later. As in the bivariate case, the prior PDF $p(\mathbf{y})$ is Gaussian, as well as the posterior PDF $p(\mathbf{y}|\mathbf{x})$. The question may arise as to when the jointly Gaussian assumption may be made. In the next section we examine an important data model for which this holds, the Bayesian linear model.

10.6 Bayesian Linear Model

Recall that in Example 10.1 the data model was

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $A \sim \mathcal{N}(\mu_A, \sigma_A^2)$, and $w[n]$ is WGN independent of A. In terms of vectors we have the equivalent data model

$$\mathbf{x} = \mathbf{1}A + \mathbf{w}.$$

This appears to be of the form of the linear model described in Chapter 4 except for the assumption that A is a random variable. It should not then be surprising that a Bayesian equivalent of the general linear model can be defined. In particular let the data be modeled as

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \quad (10.26)$$

where $\mathbf{x}$ is an $N \times 1$ data vector, $\mathbf{H}$ is a known $N \times p$ matrix, $\boldsymbol{\theta}$ is a $p \times 1$ random vector with prior PDF $\mathcal{N}(\boldsymbol{\mu}_\theta, \mathbf{C}_\theta)$, and $\mathbf{w}$ is an $N \times 1$ noise vector with PDF $\mathcal{N}(\mathbf{0}, \mathbf{C}_w)$ and independent of $\boldsymbol{\theta}$. This data model is termed the *Bayesian general linear model.* It differs from the classical general linear model in that $\boldsymbol{\theta}$ is modeled as a random variable with a Gaussian prior PDF. It will be of interest in deriving Bayesian estimators to have an explicit expression for the posterior PDF $p(\boldsymbol{\theta}|\mathbf{x})$. From Theorem 10.2 we know that if $\mathbf{x}$ and $\boldsymbol{\theta}$ are jointly Gaussian, then the posterior PDF is also Gaussian. Hence, it only remains to verify that this is indeed the case. Let $\mathbf{z} = [\mathbf{x}^T \; \boldsymbol{\theta}^T]^T$, so that from (10.26) we have

$$\begin{aligned} \mathbf{z} &= \begin{bmatrix} \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \\ \boldsymbol{\theta} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{H} & \mathbf{I} \\ \mathbf{I} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{w} \end{bmatrix} \end{aligned}$$

where the identity matrices are of dimension $N \times N$ (upper right) and $p \times p$ (lower left), and $\mathbf{0}$ is an $N \times N$ matrix of zeros. Since $\boldsymbol{\theta}$ and $\mathbf{w}$ are independent of each other and each one is Gaussian, they are jointly Gaussian. Furthermore, because $\mathbf{z}$ is a linear

transformation of a Gaussian vector, it too is Gaussian. Hence, Theorem 10.2 applies directly, and we need only determine the mean and covariance of the posterior PDF. We identify $\mathbf{x}$ as $\mathbf{H}\boldsymbol{\theta} + \mathbf{w}$ and $\mathbf{y}$ as $\boldsymbol{\theta}$ to obtain the means

$$\begin{aligned} E(\mathbf{x}) &= E(\mathbf{H}\boldsymbol{\theta} + \mathbf{w}) = \mathbf{H}E(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\mu}_\theta \\ E(\mathbf{y}) &= E(\boldsymbol{\theta}) = \boldsymbol{\mu}_\theta \end{aligned}$$

and covariances

$$\begin{aligned} \mathbf{C}_{xx} &= E\left[(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^T\right] \\ &= E\left[(\mathbf{H}\boldsymbol{\theta} + \mathbf{w} - \mathbf{H}\boldsymbol{\mu}_\theta)(\mathbf{H}\boldsymbol{\theta} + \mathbf{w} - \mathbf{H}\boldsymbol{\mu}_\theta)^T\right] \\ &= E\left[(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta) + \mathbf{w})(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta) + \mathbf{w})^T\right] \\ &= \mathbf{H}E\left[(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)^T\right]\mathbf{H}^T + E(\mathbf{w}\mathbf{w}^T) \\ &= \mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w \end{aligned}$$

recalling that $\boldsymbol{\theta}$ and $\mathbf{w}$ are independent. Also, the cross-covariance matrix is

$$\begin{aligned} \mathbf{C}_{yx} &= E\left[(\mathbf{y} - E(\mathbf{y}))(\mathbf{x} - E(\mathbf{x}))^T\right] \\ &= E\left[(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta) + \mathbf{w})^T\right] \\ &= E\left[(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta)(\mathbf{H}(\boldsymbol{\theta} - \boldsymbol{\mu}_\theta))^T\right] \\ &= \mathbf{C}_\theta\mathbf{H}^T. \end{aligned}$$

We can now summarize our results for the Bayesian general linear model.

Theorem 10.3 (Posterior PDF for the Bayesian General Linear Model) *If the observed data* $\mathbf{x}$ *can be modeled as*

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{10.27}$$

where $\mathbf{x}$ *is an* $N \times 1$ *data vector,* $\mathbf{H}$ *is a known* $N \times p$ *matrix,* $\boldsymbol{\theta}$ *is a* $p \times 1$ *random vector with prior PDF* $\mathcal{N}(\boldsymbol{\mu}_\theta, \mathbf{C}_\theta)$*, and* $\mathbf{w}$ *is an* $N \times 1$ *noise vector with PDF* $\mathcal{N}(\mathbf{0}, \mathbf{C}_w)$ *and independent of* $\boldsymbol{\theta}$*, then the posterior PDF* $p(\boldsymbol{\theta}|\mathbf{x})$ *is Gaussian with mean*

$$E(\boldsymbol{\theta}|\mathbf{x}) = \boldsymbol{\mu}_\theta + \mathbf{C}_\theta\mathbf{H}^T(\mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w)^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_\theta) \tag{10.28}$$

and covariance

$$\mathbf{C}_{\theta|x} = \mathbf{C}_\theta - \mathbf{C}_\theta\mathbf{H}^T(\mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w)^{-1}\mathbf{H}\mathbf{C}_\theta. \tag{10.29}$$

In contrast to the classical general linear model, $\mathbf{H}$ need not be full rank to ensure the invertibility of $\mathbf{H}\mathbf{C}_\theta\mathbf{H}^T + \mathbf{C}_w$. We illustrate the use of these formulas by applying them to Example 10.1.

Example 10.2 - DC Level in WGN - Gaussian Prior PDF (continued)

Since $x[n] = A + w[n]$ for $n = 0, 1, \ldots, N-1$ with $A \sim \mathcal{N}(\mu_A, \sigma_A^2)$ and $w[n]$ is WGN with variance σ^2 and independent of A, we have the Bayesian general linear model

$$\mathbf{x} = \mathbf{1}A + \mathbf{w}.$$

According to Theorem 10.3, $p(A|\mathbf{x})$ is Gaussian and

$$E(A|\mathbf{x}) = \mu_A + \sigma_A^2 \mathbf{1}^T (\mathbf{1}\sigma_A^2 \mathbf{1}^T + \sigma^2 \mathbf{I})^{-1} (\mathbf{x} - \mathbf{1}\mu_A).$$

Using Woodbury's identity (see Appendix 1)

$$\left(\mathbf{I} + \frac{\sigma_A^2}{\sigma^2}\mathbf{1}\mathbf{1}^T\right)^{-1} = \mathbf{I} - \frac{\frac{\sigma_A^2}{\sigma^2}\mathbf{1}\mathbf{1}^T}{1 + N\frac{\sigma_A^2}{\sigma^2}} \tag{10.30}$$

so that

$$\begin{aligned}
E(A|\mathbf{x}) &= \mu_A + \frac{\sigma_A^2}{\sigma^2}\mathbf{1}^T \left(\mathbf{I} - \frac{\mathbf{1}\mathbf{1}^T}{N + \frac{\sigma^2}{\sigma_A^2}}\right)(\mathbf{x} - \mathbf{1}\mu_A) \\
&= \mu_A + \frac{\sigma_A^2}{\sigma^2}\left(\mathbf{1}^T - \frac{N}{N + \frac{\sigma^2}{\sigma_A^2}}\mathbf{1}^T\right)(\mathbf{x} - \mathbf{1}\mu_A) \\
&= \mu_A + \frac{\sigma_A^2}{\sigma^2}\left(1 - \frac{N}{N + \frac{\sigma^2}{\sigma_A^2}}\right)(N\bar{x} - N\mu_A) \\
&= \mu_A + \frac{N}{N + \frac{\sigma^2}{\sigma_A^2}}(\bar{x} - \mu_A) \\
&= \mu_A + \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}(\bar{x} - \mu_A).
\end{aligned} \tag{10.31}$$

It is interesting to note that in this form the MMSE estimator resembles a "sequential"-type estimator (see Section 8.7). The estimator with no data or $\hat{A} = \mu_A$ is corrected by the error between the data estimator $\bar{x}$ and the "previous" estimate μ_A. The "gain factor" $\sigma_A^2/(\sigma_A^2 + \sigma^2/N)$ depends on our confidence in the previous estimate and the current data. We will see later that a sequential MMSE estimator can be defined and will have just these properties. Finally, with some more algebra (10.11) can be obtained.

To find the posterior variance we use (10.29), so that

$$\text{var}(A|\mathbf{x}) = \sigma_A^2 - \sigma_A^2 \mathbf{1}^T (\mathbf{1}\sigma_A^2 \mathbf{1}^T + \sigma^2 \mathbf{I})^{-1} \mathbf{1}\sigma_A^2.$$

Using (10.30) once again, we obtain

$$\begin{aligned}
\text{var}(A|\mathbf{x}) &= \sigma_A^2 - \frac{\sigma_A^2}{\sigma^2}\mathbf{1}^T\left(\mathbf{I} - \frac{\mathbf{1}\mathbf{1}^T}{N + \frac{\sigma^2}{\sigma_A^2}}\right)\mathbf{1}\sigma_A^2 \\
&= \sigma_A^2 - \frac{\sigma_A^4}{\sigma^2}\left(N - \frac{N^2}{N + \frac{\sigma^2}{\sigma_A^2}}\right) \\
&= \frac{\frac{\sigma^2}{N}\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}
\end{aligned}$$

which is just (10.12). ◇

In the succeeding chapters we will make extensive use of the Bayesian linear model. For future reference we point out that the mean (10.28) and covariance (10.29) of the posterior PDF can be expressed in alternative forms as (see Problem 10.13)

$$E(\boldsymbol{\theta}|\mathbf{x}) = \boldsymbol{\mu}_\theta + \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^T\mathbf{C}_w^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\mu}_\theta) \tag{10.32}$$

and

$$\mathbf{C}_{\theta|x} = \left(\mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H}\right)^{-1} \tag{10.33}$$

or

$$\mathbf{C}_{\theta|x}^{-1} = \mathbf{C}_\theta^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H}. \tag{10.34}$$

The latter expression is particularly interesting. For the previous example we would have

$$\begin{aligned}
\frac{1}{\text{var}(A|\mathbf{x})} &= \frac{1}{\sigma_A^2} + \mathbf{1}^T(\sigma^2\mathbf{I})^{-1}\mathbf{1} \\
&= \frac{1}{\sigma_A^2} + \frac{1}{\frac{\sigma^2}{N}}.
\end{aligned}$$

This form lends itself to the interpretation that the "information" or reciprocal of the variance of the prior knowledge $1/\sigma_A^2$ and the "information" of the data $1/(\sigma^2/N)$ *add* to yield the information embodied in the posterior PDF.

10.7 Nuisance Parameters

Many estimation problems are characterized by a set of unknown parameters, of which we are really interested only in a subset. The remaining parameters, which serve only

to complicate the problem, are referred to as *nuisance* parameters. Such would be the case if for a DC level in WGN, we were interested in estimating σ^2 but A was unknown. The DC level A would be the nuisance parameter. If we assume that the parameters are deterministic, as in the classical estimation approach, then in general we have no alternative but to estimate σ^2 *and* A. In the Bayesian approach we can rid ourselves of nuisance parameters by "integrating them out." Suppose the unknown parameters to be estimated are $\boldsymbol{\theta}$ and some additional nuisance parameters $\boldsymbol{\alpha}$ are present. Then, if $p(\boldsymbol{\theta}, \boldsymbol{\alpha}|\mathbf{x})$ denotes the posterior PDF, we can determine the posterior PDF of $\boldsymbol{\theta}$ *only* as

$$p(\boldsymbol{\theta}|\mathbf{x}) = \int p(\boldsymbol{\theta}, \boldsymbol{\alpha}|\mathbf{x})\, d\boldsymbol{\alpha}. \tag{10.35}$$

We can also express this as

$$p(\boldsymbol{\theta}|\mathbf{x}) = \frac{p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})\, d\boldsymbol{\theta}} \tag{10.36}$$

where

$$p(\mathbf{x}|\boldsymbol{\theta}) = \int p(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\alpha})p(\boldsymbol{\alpha}|\boldsymbol{\theta})\, d\boldsymbol{\alpha}. \tag{10.37}$$

If we furthermore assume that the nuisance parameters are independent of the desired parameters, then (10.37) reduces to

$$p(\mathbf{x}|\boldsymbol{\theta}) = \int p(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\alpha})p(\boldsymbol{\alpha})\, d\boldsymbol{\alpha}. \tag{10.38}$$

We observe that the nuisance parameters are first integrated out of the conditional PDF $p(\mathbf{x}|\boldsymbol{\theta}, \boldsymbol{\alpha})$ and then the posterior PDF is found as usual by Bayes' theorem. If a MMSE estimator is desired, we need only determine the mean of the posterior PDF. The nuisance parameters no longer enter into the problem. Of course, their presence will affect the final estimator since from (10.38) $p(\mathbf{x}|\boldsymbol{\theta})$ depends on $p(\boldsymbol{\alpha})$. From a theoretical viewpoint, the Bayesian approach does not suffer from the problems of classical estimators in which nuisance parameters may invalidate an estimator. We now illustrate the approach with an example.

Example 10.3 - Scaled Covariance Matrix

Assume that we observe the $N \times 1$ data vector $\mathbf{x}$ whose conditional PDF $p(\mathbf{x}|\boldsymbol{\theta}, \sigma^2)$ is $\mathcal{N}(\mathbf{0}, \sigma^2\mathbf{C}(\boldsymbol{\theta}))$. (The reader should not confuse $\mathbf{C}(\boldsymbol{\theta})$, the scaled covariance matrix of $\mathbf{x}$, with $\mathbf{C}_\theta$, the covariance matrix of $\boldsymbol{\theta}$.) The parameter $\boldsymbol{\theta}$ is to be estimated, and σ^2 is to be regarded as a nuisance parameter. The covariance matrix depends on $\boldsymbol{\theta}$ in some unspecified manner. We assign the prior PDF to σ^2 of

$$p(\sigma^2) = \begin{cases} \dfrac{\lambda \exp\left(-\frac{\lambda}{\sigma^2}\right)}{\sigma^4} & \sigma^2 > 0 \\ 0 & \sigma^2 < 0 \end{cases} \tag{10.39}$$

where $\lambda > 0$, and assume σ^2 to be independent of $\boldsymbol{\theta}$. The prior PDF is a special case of the *inverted gamma* PDF. Then, from (10.38) we have

$$\begin{aligned} p(\mathbf{x}|\boldsymbol{\theta}) &= \int p(\mathbf{x}|\boldsymbol{\theta},\sigma^2)p(\sigma^2)\,d\sigma^2 \\ &= \int_0^\infty \frac{1}{(2\pi)^{\frac{N}{2}}\det^{\frac{1}{2}}[\sigma^2\mathbf{C}(\boldsymbol{\theta})]}\exp\left[-\frac{1}{2}\mathbf{x}^T(\sigma^2\mathbf{C}(\boldsymbol{\theta}))^{-1}\mathbf{x}\right]\frac{\lambda\exp\left(-\frac{\lambda}{\sigma^2}\right)}{\sigma^4}\,d\sigma^2 \\ &= \int_0^\infty \frac{1}{(2\pi)^{\frac{N}{2}}\sigma^N\det^{\frac{1}{2}}[\mathbf{C}(\boldsymbol{\theta})]}\exp\left[-\frac{1}{2\sigma^2}\mathbf{x}^T\mathbf{C}^{-1}(\boldsymbol{\theta})\mathbf{x}\right]\frac{\lambda\exp\left(-\frac{\lambda}{\sigma^2}\right)}{\sigma^4}\,d\sigma^2. \end{aligned}$$

Letting $\xi = 1/\sigma^2$, we have

$$p(\mathbf{x}|\boldsymbol{\theta}) = \frac{\lambda}{(2\pi)^{\frac{N}{2}}\det^{\frac{1}{2}}[\mathbf{C}(\boldsymbol{\theta})]}\int_0^\infty \xi^{\frac{N}{2}}\exp\left[-\left(\lambda+\frac{1}{2}\mathbf{x}^T\mathbf{C}^{-1}(\boldsymbol{\theta})\mathbf{x}\right)\xi\right]\,d\xi.$$

But

$$\int_0^\infty x^{m-1}\exp(-ax)\,dx = a^{-m}\Gamma(m)$$

for $a > 0$ and $m > 0$, resulting from the properties of the gamma integral. Hence, the integral can be evaluated to yield

$$p(\mathbf{x}|\boldsymbol{\theta}) = \frac{\lambda\Gamma(\frac{N}{2}+1)}{(2\pi)^{\frac{N}{2}}\det^{\frac{1}{2}}[\mathbf{C}(\boldsymbol{\theta})]\left(\lambda+\frac{1}{2}\mathbf{x}^T\mathbf{C}^{-1}(\boldsymbol{\theta})\mathbf{x}\right)^{\frac{N}{2}+1}}.$$

The posterior PDF may be found by substituting this into (10.36) (at least in theory!) ◇

10.8 Bayesian Estimation for Deterministic Parameters

Although strictly speaking the Bayesian approach can be applied only when $\boldsymbol{\theta}$ is random, in practice it is often used for deterministic parameter estimation. By this we mean that the Bayesian assumptions are made to obtain an estimator, the MMSE estimator for example, and then used as if $\boldsymbol{\theta}$ were nonrandom. Such might be the case if no MVU estimator existed. For instance, we may not be able to find an unbiased estimator that is uniformly better in terms of variance than all others (see Example 2.3). Within the Bayesian framework, however, the MMSE estimator *always exists* and thus provides an estimator which at least "on the average" (as different values of $\boldsymbol{\theta}$ are chosen) works well. Of course, for a *particular* $\boldsymbol{\theta}$ it may not perform well, and this is the risk we take by applying it for a deterministic parameter. To illustrate this potential pitfall consider Example 10.1 for which (see (10.11))

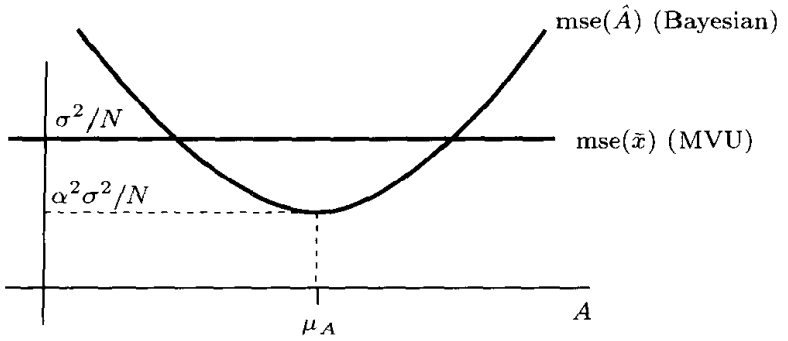


Figure 10.8 Mean square error of MVU and Bayesian estimators for deterministic DC level in WGN

$$\begin{aligned}\hat{A} &= \frac{\sigma_A^2}{\sigma_A^2+\sigma^2/N}\bar{x}+\frac{\sigma^2/N}{\sigma_A^2+\sigma^2/N}\mu_A \\ &= \alpha\bar{x}+(1-\alpha)\mu_A\end{aligned}$$

and $0<\alpha<1$. If A is a *deterministic* parameter, we can evaluate the MSE using (2.6) as

$$\text{mse}(\hat{A}) = \text{var}(\hat{A}) + b^2(\hat{A})$$

where $b(\hat{A}) = E(\hat{A}) - A$ is the bias. Then,

$$\begin{aligned}\text{mse}(\hat{A}) &= \alpha^2\text{var}(\bar{x}) + [\alpha A + (1-\alpha)\mu_A - A]^2 \\ &= \alpha^2\frac{\sigma^2}{N} + (1-\alpha)^2(A-\mu_A)^2. \end{aligned} \tag{10.40}$$

It is seen that the use of the Bayesian estimator reduces the variance, since $0<\alpha<1$, but may substantially increase the bias component of the MSE. As shown further in Figure 10.8, the Bayesian estimator exhibits less MSE than the MVU estimator $\bar{x}$ only if A is close to the prior mean μ_A. Otherwise, it is a poorer estimator. It does not have the desirable property of being *uniformly* less in MSE than all other estimators but only "on the average" or in a Bayesian sense. Hence, only if A is random and mse($\hat{A}$) can thus be considered to be the MSE *conditioned* on a known value of A, do we obtain

$$\begin{aligned}\text{Bmse}(\hat{A}) &= E_A[\text{mse}(\hat{A})] \\ &= \alpha^2\frac{\sigma^2}{N} + (1-\alpha)^2E_A[(A-\mu_A)^2] \\ &= \alpha^2\frac{\sigma^2}{N} + (1-\alpha)^2\sigma_A^2 \\ &= \frac{\sigma^2}{N}\frac{\sigma_A^2}{\sigma_A^2+\frac{\sigma^2}{N}} \\ &< \frac{\sigma^2}{N} = \text{Bmse}(\bar{x}),\end{aligned}$$

or that the *Bayesian* MSE is less. In effect, the Bayesian MMSE estimator trades off bias for variance in an attempt to reduce the overall MSE. In doing so it benefits from the prior knowledge that $A \sim \mathcal{N}(\mu_A, \sigma_A^2)$. This allows it to adjust α so that the overall MSE, on the average, is less. The large MSEs shown in Figure 10.8 for A not near μ_A are of no consequence to the choice of α since they occur infrequently. Of course, the classical estimation approach will not have this advantage, being required to produce an estimator that has good performance for *all* A.

A second concern is the choice of the prior PDF. If no prior knowledge is available, as we assume in classical estimation, then we do not want to apply a Bayesian estimator based on a highly concentrated prior PDF. Again considering the same example, if A is deterministic but we apply the Bayesian estimator anyway, we see that

$$E(\hat{A}) = \frac{\sigma_A^2}{\sigma_A^2 + \sigma^2/N} A + \frac{\sigma^2/N}{\sigma_A^2 + \sigma^2/N} \mu_A.$$

There is a bias and it is towards μ_A. To alleviate this bias we would need to specify a prior PDF that is nearly flat, or for this example, by letting $\sigma_A^2 \to \infty$. Otherwise, as already described, the MSE can be quite large for A not near μ_A. (Also, note that as $\sigma_A^2 \to \infty$, we have $\hat{A} \to \bar{x}$ and $\text{mse}(\hat{A}) \to \text{mse}(\bar{x})$, so that the two curves in Figure 10.8 become identical.) In general, such a prior PDF is termed a *noninformative* prior PDF. The use of a Bayesian estimator for a deterministic parameter is often justified on the basis that the noninformative prior PDF does not add any information to the problem. How the prior PDF is chosen for a given problem is beyond the scope of our discussions (see Problems 10.15–10.17 for one possible approach). Suffice it to say that there is a good deal of controversy surrounding this idea. The interested reader should consult [Box and Tiao 1973] for further details and philosophy.

References

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Box, G.E.P., G.C. Tiao, *Bayesian Inference in Statistical Analysis*, Addison-Wesley, Reading, Mass., 1973.

Gallager, R.G., *Information Theory and Reliable Communication*, J. Wiley, New York, 1968.

Zacks, S., *Parametric Statistical Inference*, Pergamon, New York, 1981.

Problems

10.1 In this problem we attempt to apply the Bayesian approach to the estimation of a deterministic parameter. Since the parameter is deterministic, we assign the prior PDF $p(\theta) = \delta(\theta - \theta_0)$, where θ_0 is the true value. Find the MMSE estimator for this prior PDF and explain your results.

10.2 Two random variables x and y are said to be conditionally independent of each other if the joint conditional PDF factors as

$$p(x, y|z) = p(x|z)p(y|z)$$

where z is the conditioning random variable. We observe $x[n] = A + w[n]$ for $n = 0, 1$, where A, $w[0]$, and $w[1]$ are random variables. If $A, w[0], w[1]$ are all independent, prove that $x[0]$ and $x[1]$ are conditionally independent. The conditioning random variable is A. Are $x[0]$ and $x[1]$ independent unconditionally or is it true that

$$p(x[0], x[1]) = p(x[0])p(x[1])?$$

To answer this consider the case where $A, w[0], w[1]$ are independent and each random variable has the PDF $\mathcal{N}(0, 1)$.

10.3 The data $x[n]$ for $n = 0, 1, \ldots, N-1$ are observed, each sample having the conditional PDF

$$p(x[n]|\theta) = \begin{cases} \exp\left[-(x[n] - \theta)\right] & x[n] > \theta \\ 0 & x[n] < \theta, \end{cases}$$

and conditioned on θ the observations are independent. The prior PDF is

$$p(\theta) = \begin{cases} \exp(-\theta) & \theta > 0 \\ 0 & \theta < 0. \end{cases}$$

Find the MMSE estimator of θ. Note: See Problem 10.2 for the definition of conditional independence.

10.4 Repeat Problem 10.3 but with the conditional PDF

$$p(x[n]|\theta) = \begin{cases} \frac{1}{\theta} & 0 \leq x[n] \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

and the uniform prior PDF $\theta \sim \mathcal{U}[0, \beta]$. What happens if β is very large so that there is little prior knowledge?

10.5 Rederive the MMSE estimator by letting

$$\begin{aligned} \text{Bmse}(\hat{\theta}) &= E_{x,\theta}\left[(\theta - \hat{\theta})^2\right] \\ &= E_{x,\theta}\left\{\left[(\theta - E(\theta|\mathbf{x})) + \left(E(\theta|\mathbf{x}) - \hat{\theta}\right)\right]^2\right\} \end{aligned}$$

and evaluating. Hint: Use the result $E_{x,\theta}(\,) = E_x\left[E_{\theta|x}(\,)\right]$.

10.6 In Example 10.1 modify the data model as follows:

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ_+^2 if $A \geq 0$ and σ_-^2 if $A < 0$. Find the PDF $p(\mathbf{x}|A)$. Compare this to $p(\mathbf{x}; A)$ for the classical case in which A is deterministic

and $w[n]$ is WGN with variance σ^2 for

a. $\sigma_+^2 = \sigma_-^2$

b. $\sigma_+^2 \neq \sigma_-^2$.

10.7 Plot $\hat{A}$ as given by (10.9) as a function of $\bar{x}$ if $\sqrt{\sigma^2/N} = 1$ for $A_0 = 3$ and $A_0 = 10$. Compare your results to the estimator $\hat{A} = \bar{x}$. Hint: Note that the numerator can be evaluated in closed form and the denominator is related to the cumulative distribution function of a Gaussian random variable.

10.8 A random variable θ has the PDF $p(\theta)$. It is desired to estimate a realization of θ without the availability of any data. To do so a MMSE estimator is proposed that minimizes $E[(\theta - \hat{\theta})^2]$, where the expectation is with respect to $p(\theta)$ only. Prove that the MMSE estimator is $\hat{\theta} = E(\theta)$. Apply your results to Example 10.1 to show that the minimum Bayesian MSE is reduced when data are incorporated into the estimator.

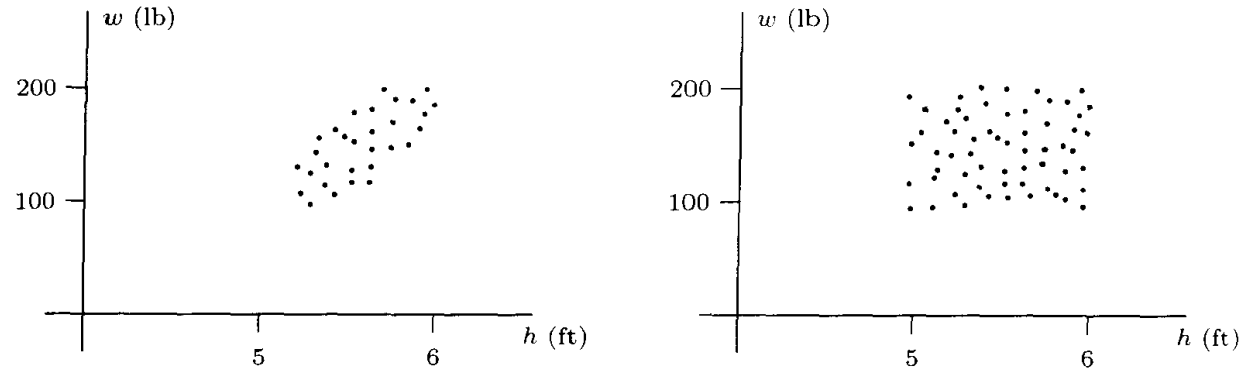
10.9 A quality assurance inspector has the job of monitoring the resistance values of manufactured resistors. He does so by choosing a resistor from a batch and measuring its resistance with an ohmmeter. He knows that the ohmmeter is of poor quality and imparts an error to the measurement which he models as a $\mathcal{N}(0,1)$ random variable. Hence, he takes N independent measurements. Also, he knows that the resistors should be 100 ohms. Due to manufacturing tolerances, however, they generally are in error by ϵ, where $\epsilon \sim \mathcal{N}(0, 0.011)$. If the inspector chooses a resistor, how many ohmmeter measurements are necessary to ensure that a MMSE estimator of the resistance R yields the correct resistance to 0.1 ohms "on the average" or as he continues to choose resistors throughout the day? How many measurements would he need if he did not have any prior knowledge about the manufacturing tolerances?

10.10 In this problem we discuss reproducing PDFs. Recall that

$$p(\theta|\mathbf{x}) = \frac{p(\mathbf{x}|\theta)p(\theta)}{\int p(\mathbf{x}|\theta)p(\theta)\,d\theta}$$

where the denominator does not depend on θ. If $p(\theta)$ is chosen so that when multiplied by $p(\mathbf{x}|\theta)$ we obtain the same form of PDF in θ, then the posterior PDF $p(\theta|\mathbf{x})$ will have the same form as $p(\theta)$. Such was the case in Example 10.1 for the Gaussian PDF. Now assume that the PDF of $x[n]$ conditioned on θ is the exponential PDF

$$p(x[n]|\theta) = \begin{cases} \theta \exp(-\theta x[n]) & x[n] > 0 \\ 0 & x[n] < 0 \end{cases}$$



(a) Data for planet Earth

(b) Data for faraway planet

Figure 10.9 Height-weight data

where the $x[n]$'s are conditionally independent (see Problem 10.2). Next, assume the gamma prior PDF

$$p(\theta) = \begin{cases} \dfrac{\lambda^\alpha}{\Gamma(\alpha)}\theta^{\alpha-1}\exp(-\lambda\theta) & \theta > 0 \\ 0 & \theta < 0 \end{cases}$$

where $\lambda > 0, \alpha > 0$, and find the posterior PDF. Compare it to the prior PDF. Such a PDF, in this case the gamma, is termed a *conjugate prior* PDF.

10.11 It is desired to estimate a person's weight based on his height. To see if this is feasible, data were taken for $N = 100$ people to generate the ordered pairs (h, w), where h denotes the height and w the weight. The data that were obtained appear as shown in Figure 10.9a. Explain how you might be able to guess someone's weight based on his height using a MMSE estimator. What modeling assumptions would you have to make about the data? Next, the same experiment was performed for people on a planet far away. The data obtained are shown in Figure 10.9b. What would the MMSE estimator of weight be now?

10.12 If $[x\, y]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$, where

$$\mathbf{C} = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$$

let $g(y) = p(x_0, y)$ for some $x = x_0$. Prove that $g(y)$ is maximized for $y = \rho x_0$. Also, show that $E(y|x_0) = \rho x_0$. Why are they the same? If $\rho = 0$, what is the MMSE estimator of y based on x?

10.13 Verify (10.32) and (10.33) by using the matrix inversion lemma. Hint: Verify (10.33) first and use it to verify (10.32).

10.14 The data

$$x[n] = Ar^n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where r is known, $w[n]$ is WGN with variance σ^2, and $A \sim \mathcal{N}(0, \sigma_A^2)$ independent of $w[n]$ are observed. Find the MMSE estimator of A as well as the minimum Bayesian MSE.

10.15 A measure of the randomness of a random variable θ is its *entropy* defined as

$$H(\theta) = E(-\ln p(\theta)) = -\int \ln p(\theta)\, p(\theta)\, d\theta.$$

If $\theta \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$, find the entropy and relate it to the concentration of the PDF. After observing the data, the entropy of the posterior PDF can be defined as

$$H(\theta|\mathbf{x}) = E(-\ln p(\theta|\mathbf{x})) = -\int\int \ln p(\theta|\mathbf{x})\, p(\mathbf{x}, \theta)\, d\mathbf{x} d\theta$$

and should be less than $H(\theta)$. Hence, a measure of the information gained by observing the data is

$$I = H(\theta) - H(\theta|\mathbf{x}).$$

Prove that $I \geq 0$. Under what conditions will $I = 0$? Hint: Express $H(\theta)$ as

$$H(\theta) = -\int\int \ln p(\theta) p(\mathbf{x}, \theta)\, d\mathbf{x}\, d\theta.$$

You will also need the inequality

$$\int \ln \frac{p_1(\mathbf{u})}{p_2(\mathbf{u})} p_1(\mathbf{u})\, d\mathbf{u} \geq 0$$

for PDFs $p_1(\mathbf{u}), p_2(\mathbf{u})$. Equality holds if and only if $p_1(\mathbf{u}) = p_2(\mathbf{u})$. (This approach is described more fully in [Zacks 1981] and relies upon the standard concept of mutual information in information theory [Ash 1965, Gallagher 1968].)

10.16 For Example 10.1 show that the information gained by observing the data is

$$I = \frac{1}{2} \ln \left(1 + \frac{\sigma_A^2}{\sigma^2/N} \right).$$

10.17 In choosing a prior PDF that is noninformative or does not assume any prior knowledge, the argument is made that we should do so to gain maximum information from the data. In this way the data are the principal contributor to our state of knowledge about the unknown parameter. Using the results of Problem 10.15, this approach may be implemented by choosing $p(\theta)$ for which I is maximum. For the Gaussian prior PDF for Example 10.1 how should μ_A and σ_A^2 be chosen so that $p(A)$ is noninformative?

Appendix 10A

Derivation of Conditional Gaussian PDF

In this appendix we derive the conditional PDF of a multivariate Gaussian PDF as summarized in Theorem 10.2. The reader will note that the result is the vector extension of that given in Example 10.1. Also, the posterior mean and covariance expressions are valid even if the covariance matrix $\mathbf{C}$ is singular. Such was the case in Appendix 7C in which $\mathbf{x}$ was a linear function of $\mathbf{y}$. The derivation in this appendix, however, assumes an invertible covariance matrix.

Using the notation of Theorem 10.2, we have

$$\begin{aligned} p(\mathbf{y}|\mathbf{x}) &= \frac{p(\mathbf{x},\mathbf{y})}{p(\mathbf{x})} \\ &= \frac{\dfrac{1}{(2\pi)^{\frac{k+l}{2}}\det^{\frac{1}{2}}(\mathbf{C})}\exp\left[-\dfrac{1}{2}\left(\begin{bmatrix}\mathbf{x}-E(\mathbf{x})\\ \mathbf{y}-E(\mathbf{y})\end{bmatrix}\right)^T\mathbf{C}^{-1}\left(\begin{bmatrix}\mathbf{x}-E(\mathbf{x})\\ \mathbf{y}-E(\mathbf{y})\end{bmatrix}\right)\right]}{\dfrac{1}{(2\pi)^{\frac{k}{2}}\det^{\frac{1}{2}}(\mathbf{C}_{xx})}\exp\left[-\dfrac{1}{2}(\mathbf{x}-E(\mathbf{x}))^T\mathbf{C}_{xx}^{-1}(\mathbf{x}-E(\mathbf{x}))\right]} \end{aligned}$$

That $p(\mathbf{x})$ can be written this way or that $\mathbf{x}$ is also multivariate Gaussian with the given mean and covariance follows from the fact that $\mathbf{x}$ and $\mathbf{y}$ are jointly Gaussian. To verify this let

$$\mathbf{x} = \begin{bmatrix}\mathbf{I} & \mathbf{0}\\ \mathbf{0} & \mathbf{0}\end{bmatrix}\begin{bmatrix}\mathbf{x}\\ \mathbf{y}\end{bmatrix}$$

and apply the property of a Gaussian vector that a linear transformation also produces a Gaussian vector. We next examine the determinant of the partitioned covariance matrix. Since the determinant of a partitioned matrix may be evaluated as

$$\det\left(\begin{bmatrix}\mathbf{A}_{11} & \mathbf{A}_{12}\\ \mathbf{A}_{21} & \mathbf{A}_{22}\end{bmatrix}\right) = \det(\mathbf{A}_{11})\det(\mathbf{A}_{22}-\mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})$$

it follows that

$$\det(\mathbf{C}) = \det(\mathbf{C}_{xx}) \det(\mathbf{C}_{yy} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy})$$

and thus

$$\frac{\det(\mathbf{C})}{\det(\mathbf{C}_{xx})} = \det(\mathbf{C}_{yy} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy}).$$

We thus have that

$$p(\mathbf{y}|\mathbf{x}) = \frac{1}{(2\pi)^{\frac{l}{2}} \det^{\frac{1}{2}}(\mathbf{C}_{yy} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy})} \exp\left(-\frac{1}{2}Q\right)$$

where

$$Q = \begin{bmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{bmatrix}^T \mathbf{C}^{-1} \begin{bmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{bmatrix} - (\mathbf{x} - E(\mathbf{x}))^T \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x})).$$

To evaluate Q we use the matrix inversion formula for a partitioned symmetric matrix

$$\begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}^{-1} =$$

$$\begin{bmatrix} (\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1} & -\mathbf{A}_{11}^{-1}\mathbf{A}_{12}(\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1} \\ -(\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1}\mathbf{A}_{21}\mathbf{A}_{11}^{-1} & (\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1} \end{bmatrix}.$$

Using this form, the off-diagonal blocks are transposes of each other so that the inverse matrix must be symmetric. This is because $\mathbf{C}$ is symmetric, and hence $\mathbf{C}^{-1}$ is symmetric. By using the matrix inversion lemma we have

$$(\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1} = \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1}\mathbf{A}_{12}(\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1}\mathbf{A}_{21}\mathbf{A}_{11}^{-1}$$

so that

$$\mathbf{C}^{-1} = \begin{bmatrix} \mathbf{C}_{xx}^{-1} + \mathbf{C}_{xx}^{-1}\mathbf{C}_{xy}\mathbf{B}^{-1}\mathbf{C}_{yx}\mathbf{C}_{xx}^{-1} & -\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy}\mathbf{B}^{-1} \\ -\mathbf{B}^{-1}\mathbf{C}_{yx}\mathbf{C}_{xx}^{-1} & \mathbf{B}^{-1} \end{bmatrix}$$

where

$$\mathbf{B} = \mathbf{C}_{yy} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy}.$$

The inverse can be written in factored form as

$$\mathbf{C}^{-1} = \begin{bmatrix} \mathbf{I} & -\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{C}_{xx}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{B}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}_{yx}\mathbf{C}_{xx}^{-1} & \mathbf{I} \end{bmatrix}$$

so that upon letting $\tilde{\mathbf{x}} = \mathbf{x} - E(\mathbf{x})$ and $\tilde{\mathbf{y}} = \mathbf{y} - E(\mathbf{y})$ we have

$$Q = \begin{bmatrix} \tilde{\mathbf{x}} \\ \tilde{\mathbf{y}} \end{bmatrix}^T \begin{bmatrix} \mathbf{I} & -\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy} \\ \mathbf{0} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{C}_{xx}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{B}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{C}_{yx}\mathbf{C}_{xx}^{-1} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}} \\ \tilde{\mathbf{y}} \end{bmatrix} - \tilde{\mathbf{x}}^T\mathbf{C}_{xx}^{-1}\tilde{\mathbf{x}}$$

$$
\begin{aligned}
&= \begin{bmatrix} \tilde{\mathbf{x}} \\ \tilde{\mathbf{y}} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\tilde{\mathbf{x}} \end{bmatrix}^T \begin{bmatrix} \mathbf{C}_{xx}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{B}^{-1} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}} \\ \tilde{\mathbf{y}} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\tilde{\mathbf{x}} \end{bmatrix} - \tilde{\mathbf{x}}^T\mathbf{C}_{xx}^{-1}\tilde{\mathbf{x}} \\
&= (\tilde{\mathbf{y}} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\tilde{\mathbf{x}})^T\mathbf{B}^{-1}(\tilde{\mathbf{y}} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\tilde{\mathbf{x}})
\end{aligned}
$$

or finally

$$
\begin{aligned}
Q = \; & \left[\mathbf{y} - (E(\mathbf{y}) + \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})))\right]^T \\
& \left[\mathbf{C}_{yy} - \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{xy}\right]^{-1} \left[\mathbf{y} - (E(\mathbf{y}) + \mathbf{C}_{yx}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})))\right].
\end{aligned}
$$

The mean of the posterior PDF is therefore given by (10.24), and the covariance by (10.25).